THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET B

Chapters 11 & 12: **The World of Algorithms, Quadrilaterals** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. By Euclid's algorithm, gcd(1001, 143) is: 1

(a) 286 **(b)** 143 **(c)** 144 **(d)** 71

2. The number of times Step 3 of Euclid's subtraction algorithm is applied while computing gcd(21, 6) is: 1

(a) 4 **(b)** 5 **(c)** 6 **(d)** 7

3. Using the long-division (Āryabhaṭa) form of Euclid's algorithm, the number of reductions needed for gcd(99, 2) is: 1

(a) 3 **(b)** 4 **(c)** 1 **(d)** 2

4. The improved gcd algorithm that keeps most-recent-common-divisor scans k from 2 up to min(54000, 81000). The number of values of k scanned is: 1

(a) 53999 **(b)** 53998 **(c)** 80999 **(d)** 54000

5. The list of common divisors of 375 and 825, in increasing order, ends with: 1

(a) 75 **(b)** 76 **(c)** 825 **(d)** 188

6. In a parallelogram ABCD, $\angle A = 110°$. Then $\angle B$ is: 1

(a) 70° **(b)** 250° **(c)** 35° **(d)** 110°

7. In $\triangle ABC$, P and Q are the midpoints of AB and AC and BC = 26 cm. The length of PQ is: 1

(a) 15 cm **(b)** 52 cm **(c)** 13 cm **(d)** 26 cm

8. The median AD of $\triangle ABC$ has length 18 cm and G is the centroid. Then AG is: 1

(a) 9 cm **(b)** 16 cm **(c)** 12 cm **(d)** 6 cm

9. G is the centroid of $\triangle ABC$ and GD = 7 cm, where AD is a median. Then AD is: 1

(a) 22 cm **(b)** 21 cm **(c)** 14 cm **(d)** 28 cm

10. The diagonals of a quadrilateral ABCD are AC = 16 cm and BD = 12 cm. The perimeter of the parallelogram formed by joining the midpoints of its sides is: 1

(a) 14 cm **(b)** 28 cm **(c)** 96 cm **(d)** 56 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Compute gcd(60, 16) by the long-division (division) algorithm. 2

12. What is 17 mod 5 and 33 mod 7? What do they represent? 2

13. Can a quadrilateral have angles $100°$, $80°$, $120°$ and $70°$? Give a reason. 2

14. Three angles of a quadrilateral are 85°, 95° and 110°. Find the fourth angle. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. Describe an algorithm to find the least common multiple (lcm) of two numbers m and n. 3

16. In $\triangle ABC$, M and N are the midpoints of AB and AC and D is any point on BC. Show that MN bisects AD. 3

17. If List 1 and List 2 are in increasing order, write an algorithm to find elements of List 1 that are not in List 2. 3

OR

Divisors occur in pairs. Write the divisors of 36 in pairs and state how many numbers need to be examined to find all divisors of 36. 3

18. Show that the sum of the angles of a convex quadrilateral is 360°. 3

OR

True or false? Justify. 3

(i) Diagonals of a rectangle are equal.

(ii) If the diagonals of a parallelogram are equal, it is a rectangle.

(iii) A quadrilateral with all sides equal is always a square.

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Use prime factorisations to compute gcd(180, 300). Show how the factorisations may be stored as lists and compared. 5

OR

Case study: A tile shop has a 360 cm × 270 cm floor to cover with square tiles of the largest possible side, with no cutting. 5

(i) Which number must the tile side be the gcd of?

(ii) Compute it using the division algorithm.

(iii) How many tiles are needed?

(iv) How many fewer reductions does the division algorithm need than the subtraction algorithm?

20. **Case study:** A craftsman's linkage forms parallelogram ABCD with AB $= 36$ cm, BC $= 20$ cm and $\angle ABC = 65^\circ$. E and F are the midpoints of AB and CD. 5

(i) Find $\angle BCD$ and $\angle ADC$.

(ii) Show that AECF is a parallelogram.

(iii) Find AE and CF.

(iv) Find the perimeter of the linkage.

OR

Case study: A triangular park ABC has sides AB $= 60$ m, BC $= 80$ m, CA $= 100$ m. Paths are laid joining the midpoints D, E, F of AB, BC, CA. 5

(i) Find the lengths of the three paths.

(ii) How many triangles are formed and what can you say about them?

(iii) Find the total length of paths.

(iv) Is $\triangle ABC$ right-angled? Give a reason.

END OF QUESTION PAPER Total: 40 Marks